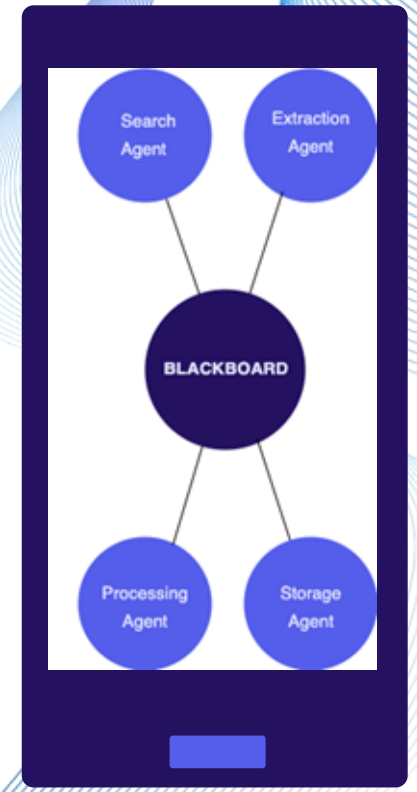
University of Essex | Online

Designing an Intelligent Multi-Agent System

for Academic Research

Blackboard-Coordinated • BDI Architecture • ArXiv API



Why Automate Academic Research Tasks?

Current Process

1. **Search**
   Find relevant papers
2. **Extract**
   Pull useful information
3. **Organise**
   Structure the results

The Problem

Time consuming

Highly repetitive

No automation

Scales poorly

The Solution

Multi-Agent System

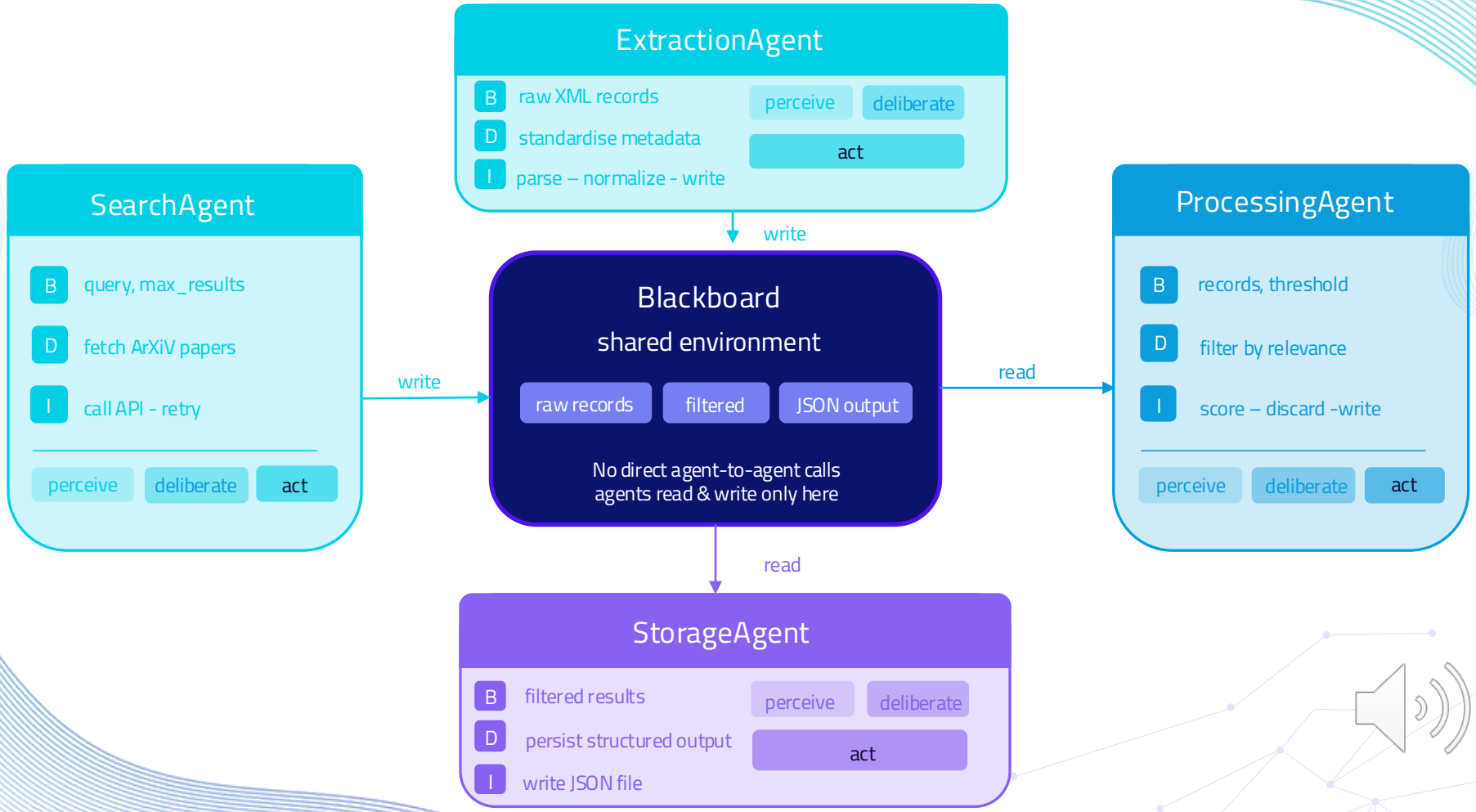
Specialised agents

Modular

Structured

Automated

From Theory to Design



Structuring the System in Code

Core Shared Modules

Defines overall behaviour – shared across all agents

Specialised Agents

Each agent in a separate module – reflects functional decomposition

Unit Tests

Validates key system behaviours – ensures reliability during development

```
BDI_MULTIAGENTSYSTEM
├── agents
│   ├── extraction_agent.py
│   ├── processing_agent.py
│   ├── search_agent.py
│   ├── storage_agent.py
├── core
│   ├── bdi_agent.py
│   ├── environment.py
│   ├── scoring.py
│   ├── utils.py
├── tests
│   ├── test_system.py
├── main.py
├── README.md
├── requirements.txt
├── results.json
```

Clarity

Modularity

Scalability

Search Agent

DATA ACCESS

Retrieves raw academic data from the ArXiv API using an HTTP GET request with a user-defined query.

INPUT → OUTPUT

IN search query

↓

OUT raw XML

BDI INTENTION

"retrieve_data"

Extraction Agent

TRANSFORMATION

Parses the raw XML response and extracts structured title and summary fields for each paper entry.

INPUT → OUTPUT

IN raw XML

↓

OUT structured records

BDI INTENTION

"extract_data"

Processing Agent

DECISION MAKING

Scores and ranks extracted records against the keyword using adaptive threshold filtering.

INPUT → OUTPUT

IN structured records

↓

OUT ranked results

BDI INTENTION

"evaluate_and_filter"

Storage Agent

DATA STORAGE

Persists the final ranked results into a JSON file, ensuring outputs are traceable and reusable.

INPUT → OUTPUT

IN ranked results

↓

OUT results.json

BDI INTENTION

"store_resul

Agents coordinate through a shared blackboard environment – no direct communication (Nii, 1986).

How the Processing Agent Decides

Scoring function

WHY ? Not all keyword matches carry equal weight

HOW ?

keyword in summary	**+2**
keyword in title	**+1**
multi-occurrence in summary	**+1**

max possible score = 4

OUTPUT A relevance score per record

feeds into →

Adaptive Filtering Loop

WHY ? The initial threshold may be too strict for some queries

HOW ?

score ≥ threshold → retain & sort (descending)

no results → threshold -= 1 → retry

while not_quality_ok AND cycle < max cycles (3)

if still empty → fallback (select best available)

Threshold progression

Cycles 2 → 1 → 0 or fallback

OUTPUT Sorted results stored in blackboard via env.update()

BDI connection – practical reasoning (Bratman, Israel and Pollack, 1988)

Seeing the System In Action

query = “machine learning”

Search Agent

```
Enter a search term: machine learning
[LOG]
[SearchAgent] --- BDI Cycle Start ---
[LOG] [SearchAgent] Preparing API request...
[LOG] [SearchAgent] Intention set: retrieve_data
[LOG] [SearchAgent] API response received successfully
[LOG] [SearchAgent] Retrieved data
[LOG] [SearchAgent] Raw data stored in environment
[LOG] [SearchAgent] --- BDI Cycle End ---
```

Extraction Agent

```
[LOG]
[ExtractionAgent] --- BDI Cycle Start ---
[LOG] [ExtractionAgent] Retrieved raw data from environment
[LOG] [ExtractionAgent] Intention set: extract_data
[LOG] [ExtractionAgent] XML parsing successful
[LOG] [ExtractionAgent] Extracted 5 records
[LOG] [ExtractionAgent] --- BDI Cycle End ---
```

Processing Agent

```
[LOG]
[ProcessingAgent] --- BDI Cycle Start ---
[LOG] [ProcessingAgent] Retrieved extracted data from environment
[LOG] [ProcessingAgent] Intention set: evaluate_and_filter
[LOG] [ProcessingAgent] Processing attempt 1
[LOG] [ProcessingAgent] Current threshold: 2
[LOG] [ProcessingAgent] Score: 4 | Title: Changing Data Sources in the Age of Mac
[LOG] [ProcessingAgent] Score: 4 | Title: DOME: Recommendations for supervised ma
[LOG] [ProcessingAgent] Score: 4 | Title: Learning Curves for Decision Making in
[LOG] [ProcessingAgent] Score: 4 | Title: Physics-Inspired Interpretability Of Ma
[LOG] [ProcessingAgent] Score: 3 | Title: Active learning for data streams: a sur
[LOG] [ProcessingAgent] Quality acceptable
[LOG] [ProcessingAgent] Processed data stored in environment
[LOG] [ProcessingAgent] --- BDI Cycle End ---
```

Storage Agent

```
[LOG]
[StorageAgent] --- BDI Cycle Start ---
[LOG] [StorageAgent] Retrieved processed data from environment
[LOG] [StorageAgent] Intention set: store_results
[LOG] [StorageAgent] Successfully stored 5 records in 'results.json'
[LOG] [StorageAgent] --- BDI Cycle End ---
```

results.json – output file


```
[
  {
    "title": "Changing Data Sources in the Age of Machine Learning for Official Statistics",
    "summary": "Data science has become increasingly essential for the production of official statistics, as it enable
  },
  {
    "title": "DOME: Recommendations for supervised machine learning validation in biology",
    "summary": "Modern biology frequently relies on machine learning to provide predictions and improve de  n proce
  },
  {
    "title": "Learning Curves for Decision Making in Supervised Machine Learning: A Survey",
    "summary": "Learning curves are a concept from social sciences that has been adopted in the context of machine lea
  },
```

Validating System Behaviour

24 total tests

7 categories

5 mocked

Category	Tests	What is validated
Environment	3	storage and retrieval
Scoring function	6	relevance signal accuracy
Processing Agent	4	adaptive filtering logic
Extraction Agent	4	XML parsing robustness
Search Agent (mocked)	4	network fault tolerance
Storage Agent	2	JSON output correctness
End-to-end pipeline	1	full pipeline integration
Total	**24**	**All passing**

```
.
----------------------------------------------------------------------
Ran 24 tests in 0.002s

OK
```

test_adaptive_threshold_lowering ProcessingAgent

```
.[LOG] [TEST] Running: test_adaptive_threshold_lowering
[LOG] [ProcessingAgent] Retrieved extracted data from environment
[LOG] [ProcessingAgent] Intention set: evaluate_and_filter
[LOG] [ProcessingAgent] Processing attempt 1
[LOG] [ProcessingAgent] Current threshold: 2
[LOG] [ProcessingAgent] Score: 0 | Title: AI Overview
[LOG] [ProcessingAgent] No results → lowering threshold to 1
[LOG] [ProcessingAgent] Processing attempt 2
[LOG] [ProcessingAgent] Current threshold: 1
[LOG] [ProcessingAgent] Score: 0 | Title: AI Overview
[LOG] [ProcessingAgent] No results → lowering threshold to 0
[LOG] [ProcessingAgent] Processing attempt 3
[LOG] [ProcessingAgent] Current threshold: 0
[LOG] [ProcessingAgent] Score: 0 | Title: AI Overview
[LOG] [ProcessingAgent] Quality acceptable
[LOG] [ProcessingAgent] Processed data stored in environment
[LOG] [TEST] ✔ Passed
```

Why this test matters

Core decision-making behaviour

Unit Testing

Validates components independently before integration (Sommerville, 2016)

What Worked – What Didn't – What Comes Next

Strengths

- Modular multi-agent design supports flexibility and maintainability
- Shared blackboard enables transparent, loosely coupled coordination
- Explicit BDI reasoning cycle ensures structured decision-making
- Unit testing with mocked APIs increases robustness and confidence

Limitations

- Keyword filtering relies on simple heuristics, limiting accuracy
- Adaptive threshold lowers quality bar rather than improving relevance
- Rule-based design cannot learn or improve from data over time
- Constrained to ArXiv only, limiting diversity of retrieved results

Future Improvements

ML FILTERING
Scikit-learn classifiers for relevance ranking

NLP & EMBEDDINGS
Semantic similarity for deeper understanding

ADAPTIVE BEHAVIOUR
Agents learn from observed outcomes

MULTI-SOURCE APIs
IEEE, Google Scholar for broader coverage

INTELLIGENT SYSTEM
From the rule-based to learned behaviour

References

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